

PDE Corpus — Final 100 Ranked Papers

Stage 4: Deep-Read & Reproduction Feasibility Assessment

REPLICATE-PROJECT, Stage 4

April 24, 2026

1 Motivation & Methodology

This report summarises the final 100 papers selected from the PDE-numerical-methods corpus (~17,000 candidates) for potential reproduction by the REPLICATE project’s PDE-solver AI agents. The pipeline was:

1. **Stage 1** — Raw corpus assembly from 7 preselected domains (Poisson, heat, Burgers, Helmholtz, Navier–Stokes, etc.), $\approx 17,000$ entries.
2. **Stage 2** — Initial heuristic scoring + top-300 shortlist.
3. **Stage 3** — Semantic Scholar enrichment (citations, venue, OA PDF URLs), domain rebalancing, top-100 shortlist.
4. **Stage 4 (this report)** — OA PDF fetch (70/100 successful), per-paper signal extraction (code repos, test problems, numerical methods, convergence tables), reproducibility/interest scoring, domain-balanced final ranking, reproduction-plan scaffolding.

1.1 Scoring Rubric

Reproducibility (1–10) combines: availability of full PDF; concrete numerical methods (FEM/FVM/FDM/DG/spectral/PINN); public code (GitHub, Zenodo, CodeOcean); compatibility with open-source stacks (FEniCS, PETSc, deal.II, OpenFOAM, Clawpack, JAX, Julia DifferentialEquations.jl); presence of convergence tables or benchmark problems; compute feasibility (< 12 h on available GPU/CPU resources).

Interest (1–10) weights: citation count, recency, methodological novelty, active research direction (ML-for-PDE, quantum algorithms), benchmark utility.

Priority: A = ready to go (repro ≥ 8 and interest ≥ 7); B = modest gaps (repro ≥ 6 and interest ≥ 5); C = hard.

1.2 Enrichment Statistics

- PDFs successfully downloaded: **70/100** (30 fell back to abstract-only due to paywall / 403).
- Topic buckets covered: **16**.
- Priority distribution: **A=14, B=45, C=41**.

2 Domain Breakdown

Topic bucket	Count
Fluid/NavierStokes	24
Elliptic	17
NonlinearScalar	14
ML-operator/PINN	9
Heat/Diffusion	9
Helmholtz/Wave	5
Hyperbolic	4
Stochastic	4
Quantum/Kinetic	4
PhaseField/RD	3
AMR/Multigrid	2
Relativity/GR	1
Fractional	1
Other	1
FreeBoundary	1
Porous/Darcy	1

3 Top 10 — Detailed Summaries

#1 — Can physics-informed neural networks beat the finite element method?

Score: 19 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2023 **Cites:** 188

Authors: T. G. Grossmann;U. J. Komorowska;J. Latz;C. Schönlieb

DOI: <https://doi.org/10.1093/imamat/hxae011>

Reproducibility (10/10): full-pdf-available; github:github.com/TamaraGrossmann/FEM-vs-PINNs.; methods:finite element,finite difference,finite volume,spectral; tools:fenics,julia,jax; tests:heat equation,poisson,navier-stokes; conv-table

Interest (9/10): cited:188; recent; active-ml-qc; good-benchmark

Core method: finite element

Test problems: heat equation, poisson, navier-stokes

Tools: PyTorch or JAX

Est. effort: 4h, GPU=True

#2 — Solving High-Dimensional PDEs with Latent Spectral Models

Score: 18 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2023 **Cites:** 95

Authors: Haixu Wu;Tengge Hu;Huakun Luo;Jianmin Wang;Mingsheng Long

DOI: <https://doi.org/10.48550/arXiv.2301.12664>

Reproducibility (10/10): full-pdf-available; github:github.com/thuml/Latent-; methods:finite element,spectral,pseudospectral, tools:pytorch; tests:navier-stokes,darcy,diffusion; conv-table

Interest (8/10): modest-cite:95; recent; active-ml-qc; good-benchmark

Core method: finite element

Test problems: navier-stokes, darcy, diffusion

Tools: PyTorch or JAX

Est. effort: 4h, GPU=True

#3 — Koopman neural operator as a mesh-free solver of non-linear partial differential equations

Score: 18 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2023 **Cites:** 74

Authors: Wei Xiong; Xiaomeng Huang; Ziyang Zhang; Ruixuan Deng; Pei Sun; Yang Tian

DOI: <https://doi.org/10.48550/arXiv.2301.10022>

Reproducibility (10/10): full-pdf-available; github:github.com/Koopman-Laboratory/; methods:finite element,finite difference,spectral,krylov; tools:pytorch; tests:burgers,navier-stokes,stokes; conv-table

Interest (8/10): modest-cite:74; recent; active-ml-qc; good-benchmark

Core method: finite element

Test problems: burgers, navier-stokes, stokes

Tools: PyTorch or JAX

Est. effort: 4h, GPU=True

#4 — Lifex-cfd: an Open-source Computational Fluid Dynamics Solver for Cardio-vascular Applications

Score: 18 **Priority:** A **Topic:** Fluid/NavierStokes **Year:** 2023 **Cites:** 61

Authors: P. C. Africa; Ivan Fumagalli; M. Bucelli; Alberto Zingaro; L. Dede'; A. Quarteroni

DOI: <https://doi.org/10.48550/arXiv.2304.12032>

Reproducibility (10/10): full-pdf-available; github:github.com/nschloe/meshio; methods:finite element,discontinuous galerkin,bdf,multigrid; tools:fenics,petsc,deal.ii; tests:poisson,navier-stokes,advection; conv-table

Interest (8/10): modest-cite:61; recent; good-benchmark

Core method: finite element

Test problems: poisson, navier-stokes, advection

Tools: FEniCS, OpenFOAM

Est. effort: 6h, GPU=False

#5 — Enhancing training of physics-informed neural networks using domain-decomposition based preconditioning strategies

Score: 18 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2023 **Cites:** 28

Authors: Alena Kopanicáková; Hardik Kothari; G. Karniadakis; Rolf Krause

DOI: <https://doi.org/10.1137/23M1583375>

Reproducibility (10/10): full-pdf-available; code-claimed; methods:finite element,multigrid,pinn,neural network; tools:pytorch; tests:burgers,allen-cahn,advection; conv-table

Interest (8/10): recent; active-ml-qc; good-benchmark

Core method: finite element

Test problems: burgers, allen-cahn, advection

Tools: PyTorch or JAX

Est. effort: 6h, GPU=True

#6 — Machine learning-accelerated computational fluid dynamics

Score: 17 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2021 **Cites:** 1123

Authors: Dmitrii Kochkov; Jamie A. Smith; Ayya Alieva; Qing Wang; M. Brenner; Stephan Hoyer

DOI: <https://doi.org/10.1073/pnas.2101784118>

Reproducibility (8/10): full-pdf-available; methods:finite element,finite difference,finite volume,spectral; tools:jax; tests:navier-stokes,advection,diffusion; conv-table; benchmark

Interest (9/10): high-cite:1123; active-ml-qc; good-benchmark

Core method: finite element
Test problems: navier-stokes, advection, diffusion
Tools: PyTorch, JAX
Est. effort: 6h, GPU=False
Risks: no public code

#7 — Dedalus: A flexible framework for numerical simulations with spectral methods

Score: 16 **Priority:** B **Topic:** Fluid/NavierStokes **Year:** 2019 **Cites:** 485

Authors: K. Burns;G. Vasil;J. Oishi;D. Lecoanet;B. Brown

DOI: <https://doi.org/10.1103/PhysRevResearch.2.023068>

Reproducibility (10/10): full-pdf-available; github:github.com/DedalusProject/methods_paper_examples; methods:finite difference,finite volume,spectral,runge-kutta; tools:fenics,firedrake,julia; tests:poisson,navier-stokes,helmholtz; benchmark

Interest (6/10): cited:485

Core method: finite difference

Test problems: poisson, navier-stokes, helmholtz

Tools: FEniCS, OpenFOAM

Est. effort: 6h, GPU=False

#8 — Physics-informed neural networks for solving Reynolds-averaged Navier-Stokes equations

Score: 16 **Priority:** A **Topic:** Fluid/NavierStokes **Year:** 2021 **Cites:** 404

Authors: Hamidreza Eivazi;Mojtaba Tahani;P. Schlatter;R. Vinuesa

DOI: <https://doi.org/10.1063/5.0095270>

Reproducibility (8/10): full-pdf-available; methods:spectral,imex,pinn,neural network; tools:tensorflow; tests:navier-stokes,stokes; benchmark

Interest (8/10): cited:404; active-ml-qc

Core method: spectral

Test problems: navier-stokes, stokes

Tools: PyTorch or JAX

Est. effort: 8h, GPU=True

Risks: no public code

#9 — Learning in Modal Space: Solving Time-Dependent Stochastic PDEs Using Physics-Informed Neural Networks

Score: 16 **Priority:** A **Topic:** ML-operator/PINN **Year:** 2019 **Cites:** 274

Authors: Dongkun Zhang;Ling Guo;G. Karniadakis

DOI: <https://doi.org/10.1137/19m1260141>

Reproducibility (8/10): full-pdf-available; methods:spectral,monte carlo,pinn,neural network; tools:tensorflow; tests:burgers,navier-stokes,advection; conv-table; benchmark

Interest (8/10): cited:274; active-ml-qc; good-benchmark

Core method: spectral

Test problems: burgers, navier-stokes, advection

Tools: PyTorch or JAX

Est. effort: 6h, GPU=True

Risks: no public code

#10 — Poisson Flow Generative Models

Score: 16 **Priority:** A **Topic:** Elliptic **Year:** 2022 **Cites:** 115

Authors: Yilun Xu;Ziming Liu;M. Tegmark;T. Jaakkola

DOI: <https://doi.org/10.48550/arXiv.2209.11178>

Reproducibility (8/10): full-pdf-available; github:github.com/Newbeeer/poisson_flow.; methods:spectral,runge-kutta,neural network; tests:poisson,diffusion

Interest (8/10): cited:115; recent

Core method: spectral

Test problems: poisson, diffusion

Tools: FEniCS, PETSc

Est. effort: 2h, GPU=False

4 Full 100 Ranked Table

Rk	Sc	Title	Pr	Yr	Ci	Rp	In	Topic
1	19	Can physics-informed neural networks beat the finite element method?	A	2023	188	10	9	ML-operator/PINN
2	18	Solving High-Dimensional PDEs with Latent Spectral Models	A	2023	95	10	8	ML-operator/PINN
3	18	Koopman neural operator as a mesh-free solver of non-linear partial differential equations	A	2023	74	10	8	ML-operator/PINN
4	18	Lifex-cfd: an Open-source Computational Fluid Dynamics Solver for Cardiovascular Applicati...	A	2023	61	10	8	Fluid/NavierStok...
5	18	Enhancing training of physics-informed neural networks using domain-decomposition based pr...	A	2023	28	10	8	ML-operator/PINN
6	17	Machine learning-accelerated computational fluid dynamics	A	2021	1123	8	9	ML-operator/PINN
7	16	Dedalus: A flexible framework for numerical simulations with spectral methods	B	2019	485	10	6	Fluid/NavierStok...
8	16	Physics-informed neural networks for solving Reynolds-averaged Navier-Stokes equations	A	2021	404	8	8	Fluid/NavierStok...
9	16	Learning in Modal Space: Solving Time-Dependent Stochastic PDEs Using Physics-Informed Neu...	A	2019	274	8	8	ML-operator/PINN
10	16	Poisson Flow Generative Models	A	2022	115	8	8	Elliptic
11	16	LNO: Laplace Neural Operator for Solving Differential Equations	A	2023	56	8	8	ML-operator/PINN
12	16	Walk on Stars: A Grid-Free Monte Carlo Method for PDEs with Neumann Boundary Conditions	A	2023	48	8	8	Elliptic
13	16	Fast Poisson solvers for spectral methods	B	2017	47	10	6	Elliptic
14	16	Deep Reinforcement Learning for Adaptive Mesh Refinement	A	2022	35	8	8	AMR/Multigrid
15	16	Godunov loss functions for modelling of hyperbolic conservation laws	A	2024	5	8	8	Hyperbolic
16	15	Plane Wave Discontinuous Galerkin Methods for the 2D Helmholtz Equation: Analysis of the p...	B	2011	199	7	8	Helmholtz/Wave
17	15	Galerkin Approximations for the Stochastic Burgers Equation	B	2013	105	7	8	NonlinearScalar
18	15	FLUPS: A Fourier-Based Library of Unbounded Poisson Solvers	B	2020	18	9	6	Elliptic
19	15	Kernel-based active subspaces with application to computational fluid dynamics parametric ...	B	2020	15	10	5	Hyperbolic
20	15	On the convergence of discontinuous Galerkin/Hermite spectral methods for the Vlasov-Poiss...	A	2022	6	8	7	Elliptic
21	14	Improvements to the APBS biomolecular solvation software suite	B	2017	2081	6	8	Elliptic
22	14	Fast Nonsymmetric Iterations and Preconditioning for Navier-Stokes Equations	B	1996	258	8	6	Fluid/NavierStok...
23	14	New Laplace and Helmholtz solvers	B	2019	51	8	6	Helmholtz/Wave
24	14	Stability and error analysis of a class of high-order IMEX schemes for Navier-stokes equat...	B	2021	43	8	6	Fluid/NavierStok...
25	14	Rate optimality of adaptive finite element methods with respect to the overall computation...	B	2020	34	8	6	Elliptic
26	14	Comparison of Adaptive Multiresolution and Adaptive Mesh Refinement Applied to Simulations...	B	2015	33	8	6	AMR/Multigrid
27	14	Kinetic.jl: A portable finite volume toolbox for scientific and neural computing	B	2021	17	8	6	ML-operator/PINN
28	14	Highly efficient and energy dissipative schemes for the time fractional Allen-Cahn equatio...	B	2021	17	8	6	PhaseField/RD
29	14	Modified Poisson-Nernst-Planck Model with Coulomb and Hard-sphere Correlations	B	2020	15	8	6	Elliptic

Rk	Sc	Title	Pr	Yr	Ci	Rp	In	Topic
30	13	Optimized Schwarz Methods without Overlap for the Helmholtz Equation	B	2002	403	7	6	Helmholtz/Wave
31	13	Space-Time Continuous Analysis of Waveform Relaxation for the Heat Equation	B	1998	208	7	6	Heat/Diffusion
32	13	Variational quantum algorithm for the Poisson equation	C	2020	108	5	8	Elliptic
33	13	A fully discrete approximation of the one-dimensional stochastic heat equation	B	2015	90	7	6	Stochastic
34	13	Numerical and exact solutions for time fractional Burgers' equation	B	2017	66	7	6	NonlinearScalar
35	13	Variational quantum algorithm based on the minimum potential energy for solving the Poisson...	B	2021	62	7	6	Elliptic
36	13	Learning Poisson Systems and Trajectories of Autonomous Systems via Poisson Neural Network...	B	2020	60	6	7	Elliptic
37	13	A geometric multiscale model for the numerical simulation of blood flow in the human left ...	B	2021	48	8	5	Fluid/NavierStok...
38	13	Lessons for adaptive mesh refinement in numerical relativity	B	2021	47	7	6	Fluid/NavierStok...
39	13	Analog quantum simulation of partial differential equations	B	2023	47	6	7	Quantum/Kinetic
40	13	Gmunu: toward multigrid based Einstein field equations solver for general-relativistic hyd...	B	2020	21	8	5	Relativity/GR
41	13	A Multivariate Spline based Collocation Method for Numerical Solution of Partial Different...	B	2021	11	7	6	Elliptic
42	13	Retrieval of Timewise Coefficients in the Heat Equation from Nonlocal Overdetermination Co...	B	2022	9	7	6	Heat/Diffusion
43	13	Numerical Solution to Recover Time-dependent Coefficient and Free Boundary from Nonlocal a...	B	2021	8	7	6	Heat/Diffusion
44	13	WaveTrain: A Python package for numerical quantum mechanics of chain-like systems based on...	B	2023	2	8	5	Quantum/Kinetic
45	12	Modelica Buildings library	C	2014	769	4	8	Fluid/NavierStok...
46	12	Numerical methods for nonlocal and fractional models	B	2020	200	6	6	Fractional
47	12	Action Principle for Newtonian Gravity.	B	2018	144	6	6	Fluid/NavierStok...
48	12	Dielectric continuum methods for quantum chemistry	B	2021	141	6	6	Quantum/Kinetic
49	12	Operator splitting for partial differential equations with Burgers nonlinearity	B	2011	109	6	6	NonlinearScalar
50	12	A Low-Rank Projector-Splitting Integrator for the Vlasov-Poisson Equation	B	2018	103	6	6	Elliptic
51	12	Characterization of Radially Symmetric Finite Time Blowup in Multidimensional Aggregation...	B	2012	60	6	6	Fluid/NavierStok...
52	12	New closed form solutions of some nonlinear pseudo-parabolic models via a new extended dir...	B	2023	49	6	6	NonlinearScalar
53	12	Error estimate of a decoupled numerical scheme for the Cahn-Hilliard-Stokes-Darcy system	B	2021	41	6	6	PhaseField/RD
54	12	A Time-Stepping DPG Scheme for the Heat Equation	B	2016	24	7	5	Heat/Diffusion
55	12	Depth analysis of variational quantum algorithms for the heat equation	B	2022	21	6	6	Quantum/Kinetic
56	12	A Review on Burgers' Equations and It's Applications	B	2023	8	6	6	NonlinearScalar
57	12	NodePy: A package for the analysis of numerical ODE solvers	C	2020	8	8	4	Other
58	12	Numerical Solution for Two-Sided Stefan Problem	B	2020	7	6	6	FreeBoundary
59	12	An Application of Computational Fluid Dynamics to Optimize Municipal Sewage Networks; A Ca...	C	2020	4	8	4	Fluid/NavierStok...
60	11	The well-posedness of the Kuramoto-Sivashinsky equation	C	1986	149	5	6	NonlinearScalar
61	11	Symmetry preserving numerical schemes for partial differential equations and their numeric...	B	2011	47	6	5	NonlinearScalar
62	11	The Wave Breaking for Whitham-Type Equations Revisited	C	2020	33	5	6	NonlinearScalar
63	11	Discrete Maximum principle of a high order finite difference scheme for a generalized Alle...	B	2021	28	6	5	PhaseField/RD
64	11	Assorted exact explicit solutions for the generalized Atangana's fractional BBM-Burgers eq...	B	2022	19	6	5	NonlinearScalar
65	11	Nonlinear stability of planar viscous shock wave to three-dimensional compressible Navier-...	B	2022	17	6	5	Fluid/NavierStok...
66	10	Heat and mass transfer	C	2025	1552	2	8	Fluid/NavierStok...
67	10	Newton-GMRES Preconditioning for Discontinuous Galerkin Discretizations of the Navier-Sto...	C	2008	259	4	6	Hyperbolic
68	10	Non-uniqueness of Leray solutions of the forced Navier-Stokes equations	C	2021	153	4	6	Fluid/NavierStok...
69	10	Multiplicative stochastic heat equations on the whole space	C	2015	138	4	6	Stochastic
70	10	A new technology for solving diffusion and heat equations	C	2017	106	3	7	Heat/Diffusion
71	10	Stochastic heat equation with rough dependence in space	C	2015	88	4	6	Stochastic
72	10	A Comparative Study of the Explicit Finite Difference Method and Physics-Informed Neural N...	C	2023	29	4	6	ML-operator/PINN
73	10	On second-order and fourth-order elliptic systems consisting of bulk and surface PDEs: Wel...	C	2020	23	6	4	Elliptic
74	10	ISWFOam: A numerical model for internal solitary wave simulation in continuously stratified...	C	2021	21	6	4	Fluid/NavierStok...
75	9	Is the Pollution Effect of the FEM Avoidable for the Helmholtz Equation Considering High W...	C	1997	703	3	6	Helmholtz/Wave
76	9	Global Solutions of the Navier-Stokes Equations for Multidimensional Compressible Flow wit...	C	1995	589	3	6	Fluid/NavierStok...

Rk	Sc	Title	Pr	Yr	Ci	Rp	In	Topic
77	9	On splitting methods for Schrödinger-Poisson and cubic nonlinear Schrödinger equations	C	2008	339	3	6	Elliptic
78	9	A locally conservative LDG method for the incompressible Navier-Stokes equations	C	2004	329	4	5	Fluid/NavierStok...
79	9	Krylov methods for the incompressible Navier-Stokes equations	C	1994	263	4	5	Fluid/NavierStok...
80	9	Feedback Boundary Stabilization of the Two-Dimensional Navier-Stokes Equations	C	2006	149	3	6	Fluid/NavierStok...
81	9	Spectral discretization of Darcy's equations coupled with the heat equation	C	2016	36	4	5	Heat/Diffusion
82	9	An integrated stairwise adaptive finite point scheme for the two-dimensional coupled Burge...	C	2024	4	3	6	NonlinearScalar
83	8	Hardy Inequalities and Some Critical Elliptic and Parabolic Problems	C	1998	577	2	6	Elliptic
84	8	Critical Exponent for a Nonlinear Wave Equation with Damping	C	2000	451	3	5	Helmholtz/Wave
85	8	On error estimates of the projection methods for the Navier-Stokes equations: Second-order...	C	1996	277	3	5	Fluid/NavierStok...
86	8	Conformable heat equation on a radial symmetric plate	C	2017	35	2	6	Heat/Diffusion
87	8	Numerical Simulation of Turbulent Flow and Pollutant Dispersion in Urban Street Canyons	C	2019	26	4	4	Fluid/NavierStok...
88	8	Fluctuations of two-dimensional stochastic heat equation and KPZ equation in subcritical r...	C	2021	16	4	4	Stochastic
89	8	Numerical simulation of flow over buildings using Open-FOAM®	C	2019	10	4	4	Fluid/NavierStok...
90	8	3D Modeling and Mechanism Analysis of Breaking Wave-Induced Seabed Scour around Monopile	C	2020	9	4	4	Fluid/NavierStok...
91	8	Numerical Simulation of Non-Darcy Flow in Naturally Fractured Tight Gas Reservoirs for Enh...	C	2024	5	3	5	Porous/Darcy
92	8	A 2D Model for 3D Periodic Deep-Water Waves	C	2022	5	2	6	Elliptic
93	8	Accuracy Verification of a 2D Adaptive Mesh Refinement Method Using Backward-Facing Step F...	C	2020	3	4	4	Fluid/NavierStok...
94	7	Remarkable statistical behavior for truncated Burgers-Hopf dynamics.	C	2000	115	2	5	NonlinearScalar
95	7	Off-fault plasticity in three-dimensional dynamic rupture simulations using a modal Discon...	C	2018	86	2	5	Hyperbolic
96	7	Minimal controllability time for the heat equation under unilateral state or control const...	C	2017	54	2	5	Heat/Diffusion
97	7	A Crank-Nicolson Approximation for the time Fractional Burgers Equation	C	2020	29	3	4	NonlinearScalar
98	7	Solving one- and two-dimensional unsteady Burgers' equation using fully implicit finite di...	C	2019	27	3	4	NonlinearScalar
99	7	Simultaneous Identification of Thermal Conductivity and Heat Source in the Heat Equation	C	2021	11	3	4	Heat/Diffusion
100	6	Fractal Burgers Equations	C	1998	222	1	5	NonlinearScalar

5 Priority A — Ready to Go

#1 [19] — Can physics-informed neural networks beat the finite element method?

ML-operator/PINN, 2023, 188 cites. Method: finite element; Tools: PyTorch or JAX; Est: 4h.

#2 [18] — Solving High-Dimensional PDEs with Latent Spectral Models

ML-operator/PINN, 2023, 95 cites. Method: finite element; Tools: PyTorch or JAX; Est: 4h.

#3 [18] — Koopman neural operator as a mesh-free solver of non-linear partial differential equations

ML-operator/PINN, 2023, 74 cites. Method: finite element; Tools: PyTorch or JAX; Est: 4h.

#4 [18] — Lifex-cfd: an Open-source Computational Fluid Dynamics Solver for Cardiovascular Applications

Fluid/NavierStokes, 2023, 61 cites. Method: finite element; Tools: FEniCS, OpenFOAM; Est: 6h.

#5 [18] — Enhancing training of physics-informed neural networks using domain-decomposition based precondition...

ML-operator/PINN, 2023, 28 cites. Method: finite element; Tools: PyTorch or JAX; Est: 6h.

#6 [17] — Machine learning-accelerated computational fluid dynamics

ML-operator/PINN, 2021, 1123 cites. Method: finite element; Tools: PyTorch, JAX; Est: 6h. Risks: no public code.

#8 [16] — Physics-informed neural networks for solving Reynolds-averaged Navier-Stokes equations

Fluid/NavierStokes, 2021, 404 cites. Method: spectral; Tools: PyTorch or JAX; Est: 8h. Risks: no public code.

- #9 [16]** — Learning in Modal Space: Solving Time-Dependent Stochastic PDEs Using Physics-Informed Neural Network. . .
ML-operator/PINN, 2019, 274 cites. Method: spectral; Tools: PyTorch or JAX; Est: 6h. Risks: no public code.
- #10 [16]** — Poisson Flow Generative Models
Elliptic, 2022, 115 cites. Method: spectral; Tools: FEniCS, PETSc; Est: 2h.
- #11 [16]** — LNO: Laplace Neural Operator for Solving Differential Equations
ML-operator/PINN, 2023, 56 cites. Method: neural network; Tools: PyTorch or JAX; Est: 6h. Risks: no public code.
- #12 [16]** — Walk on Stars: A Grid-Free Monte Carlo Method for PDEs with Neumann Boundary Conditions
Elliptic, 2023, 48 cites. Method: finite element; Tools: FEniCS, PETSc; Est: 3h. Risks: no public code.
- #14 [16]** — Deep Reinforcement Learning for Adaptive Mesh Refinement
AMR/Multigrid, 2022, 35 cites. Method: finite element; Tools: deal.II, AMReX; Est: 8h. Risks: no public code.
- #15 [16]** — Godunov loss functions for modelling of hyperbolic conservation laws
Hyperbolic, 2024, 5 cites. Method: finite difference; Tools: PyTorch or JAX; Est: 5h. Risks: no public code.
- #20 [15]** — On the convergence of discontinuous Galerkin/Hermite spectral methods for the Vlasov-Poisson system
Elliptic, 2022, 6 cites. Method: finite element; Tools: FEniCS, PETSc; Est: 3h. Risks: no public code.

6 Priority B — Needs Minor Work

- #7 [16]** Dedalus: A flexible framework for numerical simulations with spectral methods — Fluid/NavierStokes, 6h.
- #13 [16]** Fast Poisson solvers for spectral methods — Elliptic, 2h.
- #16 [15]** Plane Wave Discontinuous Galerkin Methods for the 2D Helmholtz Equation: Analysis of the p... — Helmholtz/Wave, 4h.
- #17 [15]** Galerkin Approximations for the Stochastic Burgers Equation — NonlinearScalar, 3h.
- #18 [15]** FLUPS: A Fourier-Based Library of Unbounded Poisson Solvers — Elliptic, 2h.
- #19 [15]** Kernel-based active subspaces with application to computational fluid dynamics parametric ... — Hyperbolic, 3h.
- #21 [14]** Improvements to the APBS biomolecular solvation software suite — Elliptic, 3h.
- #22 [14]** Fast Nonsymmetric Iterations and Preconditioning for Navier-Stokes Equations — Fluid/NavierStokes, 8h.
- #23 [14]** New Laplace and Helmholtz solvers — Helmholtz/Wave, 4h.
- #24 [14]** Stability and error analysis of a class of high-order IMEX schemes for Navier-stokes equat... — Fluid/NavierStokes, 8h.
- #25 [14]** Rate optimality of adaptive finite element methods with respect to the overall computation. . . — Elliptic, 3h.
- #26 [14]** Comparison of Adaptive Multiresolution and Adaptive Mesh Refinement Applied to Simulations... — AMR/Multigrid, 6h.
- #27 [14]** Kinetic.jl: A portable finite volume toolbox for scientific and neural computing — ML-operator/PINN, 6h.
- #28 [14]** Highly efficient and energy dissipative schemes for the time fractional Allen-Cahn equatio... — Phase-Field/RD, 5h.
- #29 [14]** Modified Poisson-Nernst-Planck Model with Coulomb and Hard-sphere Correlations — Elliptic, 3h.
- #30 [13]** Optimized Schwarz Methods without Overlap for the Helmholtz Equation — Helmholtz/Wave, 4h.
- #31 [13]** Space-Time Continuous Analysis of Waveform Relaxation for the Heat Equation — Heat/Diffusion, 3h.
- #33 [13]** A fully discrete approximation of the one-dimensional stochastic heat equation — Stochastic, 4h.
- #34 [13]** Numerical and exact solutions for time fractional Burgers' equation — NonlinearScalar, 3h.

- #35 [13] Variational quantum algorithm based on the minimum potential energy for solving the Poisso... — Elliptic, 2h.
- #36 [13] Learning Poisson Systems and Trajectories of Autonomous Systems via Poisson Neural Network... — Elliptic, 3h.
- #37 [13] A geometric multiscale model for the numerical simulation of blood flow in the human left ... — Fluid/NavierStokes, 8h.
- #38 [13] Lessons for adaptive mesh refinement in numerical relativity — Fluid/NavierStokes, 8h.
- #39 [13] Analog quantum simulation of partial differential equations — Quantum/Kinetic, 6h.
- #40 [13] Gmunu: toward multigrid based Einstein field equations solver for general-relativistic hyd... — Relativity/GR, 12h.
- #41 [13] A Multivariate Spline based Collocation Method for Numerical Solution of Partial Different... — Elliptic, 3h.
- #42 [13] Retrieval of Timewise Coefficients in the Heat Equation from Nonlocal Overdetermination Co... — Heat/Diffusion, 3h.
- #43 [13] Numerical Solution to Recover Time-dependent Coefficient and Free Boundary from Nonlocal a... — Heat/Diffusion, 3h.
- #44 [13] WaveTrain: A Python package for numerical quantum mechanics of chain-like systems based on... — Quantum/Kinetic, 4h.
- #46 [12] Numerical methods for nonlocal and fractional models — Fractional, 4h.
- #47 [12] Action Principle for Newtonian Gravity. — Fluid/NavierStokes, 8h.
- #48 [12] Dielectric continuum methods for quantum chemistry — Quantum/Kinetic, 6h.
- #49 [12] Operator splitting for partial differential equations with Burgers nonlinearity — NonlinearScalar, 3h.
- #50 [12] A Low-Rank Projector-Splitting Integrator for the Vlasov-Poisson Equation — Elliptic, 3h.
- #51 [12] Characterization of Radially Symmetric Finite Time Blowup in Multidimensional Aggregation ... — Fluid/NavierStokes, 8h.
- #52 [12] New closed form solutions of some nonlinear pseudo-parabolic models via a new extended dir... — NonlinearScalar, 3h.
- #53 [12] Error estimate of a decoupled numerical scheme for the Cahn-Hilliard-Stokes-Darcy system — Phase-Field/RD, 5h.
- #54 [12] A Time-Stepping DPG Scheme for the Heat Equation — Heat/Diffusion, 3h.
- #55 [12] Depth analysis of variational quantum algorithms for the heat equation — Quantum/Kinetic, 6h.
- #56 [12] A Review on Burgers' Equations and It's Applications — NonlinearScalar, 3h.
- #58 [12] Numerical Solution for Two-Sided Stefan Problem — FreeBoundary, 5h.
- #61 [11] Symmetry preserving numerical schemes for partial differential equations and their numeric... — NonlinearScalar, 3h.
- #63 [11] Discrete Maximum principle of a high order finite difference scheme for a generalized Alle... — Phase-Field/RD, 5h.
- #64 [11] Assorted exact explicit solutions for the generalized Atangana's fractional BBM-Burgers eq... — NonlinearScalar, 3h.
- #65 [11] Nonlinear stability of planar viscous shock wave to three-dimensional compressible Navier-... — Fluid/NavierStokes, 8h.

7 Priority C — Hard (defer or skip)

- #32 [13] Variational quantum algorithm for the Poisson equation — Elliptic.
- #45 [12] Modelica Buildings library — Fluid/NavierStokes.

#57 [12] NodePy: A package for the analysis of numerical ODE solvers — Other.

#59 [12] An Application of Computational Fluid Dynamics to Optimize Municipal Sewage Networks; A Ca... — Fluid/NavierStokes.

#60 [11] The well-posedness of the Kuramoto-Sivashinsky equation — NonlinearScalar.

#62 [11] The Wave Breaking for Whitham-Type Equations Revisited — NonlinearScalar.

#66 [10] Heat and mass transfer — Fluid/NavierStokes.

#67 [10] Newton-GMRES Preconditioning for Discontinuous Galerkin Discretizations of the Navier-Sto... — Hyperbolic.

#68 [10] Non-uniqueness of Leray solutions of the forced Navier-Stokes equations — Fluid/NavierStokes.

#69 [10] Multiplicative stochastic heat equations on the whole space — Stochastic.

#70 [10] A new technology for solving diffusion and heat equations — Heat/Diffusion.

#71 [10] Stochastic heat equation with rough dependence in space — Stochastic.

#72 [10] A Comparative Study of the Explicit Finite Difference Method and Physics-Informed Neural N... — ML-operator/PINN.

#73 [10] On second-order and fourth-order elliptic systems consisting of bulk and surface PDEs: Wel... — Elliptic.

#74 [10] ISWfoam: A numerical model for internal solitary wave simulation in continuously stratifie... — Fluid/NavierStokes.

#75 [9] Is the Pollution Effect of the FEM Avoidable for the Helmholtz Equation Considering High W... — Helmholtz/Wave.

#76 [9] Global Solutions of the Navier-Stokes Equations for Multidimensional Compressible Flow wit... — Fluid/NavierStokes.

#77 [9] On splitting methods for Schrödinger-Poisson and cubic nonlinear Schrödinger equations — Elliptic.

#78 [9] A locally conservative LDG method for the incompressible Navier-Stokes equations — Fluid/NavierStokes.

#79 [9] Krylov methods for the incompressible Navier-Stokes equations — Fluid/NavierStokes.

#80 [9] Feedback Boundary Stabilization of the Two-Dimensional Navier-Stokes Equations — Fluid/NavierStokes.

#81 [9] Spectral discretization of Darcy's equations coupled with the heat equation — Heat/Diffusion.

#82 [9] An integrated stairwise adaptive finite point scheme for the two-dimensional coupled Burge... — NonlinearScalar.

#83 [8] Hardy Inequalities and Some Critical Elliptic and Parabolic Problems — Elliptic.

#84 [8] Critical Exponent for a Nonlinear Wave Equation with Damping — Helmholtz/Wave.

#85 [8] On error estimates of the projection methods for the Navier-Stokes equations: Second-order... — Fluid/NavierStokes.

#86 [8] Conformable heat equation on a radial symmetric plate — Heat/Diffusion.

#87 [8] Numerical Simulation of Turbulent Flow and Pollutant Dispersion in Urban Street Canyons — Fluid/NavierStokes.

#88 [8] Fluctuations of two-dimensional stochastic heat equation and KPZ equation in subcritical r... — Stochastic.

#89 [8] Numerical simulation of flow over buildings using OpenFOAM® — Fluid/NavierStokes.

#90 [8] 3D Modeling and Mechanism Analysis of Breaking Wave-Induced Seabed Scour around Monopile — Fluid/NavierStokes.

#91 [8] Numerical Simulation of Non-Darcy Flow in Naturally Fractured Tight Gas Reservoirs for Enh... — Porous/Darcy.

#92 [8] A 2D Model for 3D Periodic Deep-Water Waves — Elliptic.

#93 [8] Accuracy Verification of a 2D Adaptive Mesh Refinement Method Using Backward-Facing Step F... — Fluid/NavierStokes.

#94 [7] Remarkable statistical behavior for truncated Burgers-Hopf dynamics. — NonlinearScalar.

#95 [7] Off-fault plasticity in three-dimensional dynamic rupture simulations using a modal Discon... — Hyperbolic.

#96 [7] Minimal controllability time for the heat equation under unilateral state or control const... — Heat/Diffusion.

#97 [7] A Crank-Nicolson Approximation for the time Fractional Burgers Equation — NonlinearScalar.

#98 [7] Solving one- and two-dimensional unsteady Burgers' equation using fully implicit finite di... — NonlinearScalar.

#99 [7] Simultaneous Identification of Thermal Conductivity and Heat Source in the Heat Equation — Heat/Diffusion.

#100 [6] Fractal Burgers Equations — NonlinearScalar.

8 Appendix: Per-paper Reproduction Details (Top 30)

For each of the top-30 ranked papers we list the core method, test problems, tool stack, GPU requirement, and estimated effort.

#1 — Can physics-informed neural networks beat the finite element method? Method: finite element. Tests: heat equation, poisson, navier-stokes. Tools: PyTorch or JAX. Est.: 4h; GPU=True. Expected fidelity: quantitative match on reported error tables.

#2 — Solving High-Dimensional PDEs with Latent Spectral Models Method: finite element. Tests: navier-stokes, darcy, diffusion. Tools: PyTorch or JAX. Est.: 4h; GPU=True. Expected fidelity: quantitative match on reported error tables.

#3 — Koopman neural operator as a mesh-free solver of non-linear partial differential equations Method: finite element. Tests: burgers, navier-stokes, stokes. Tools: PyTorch or JAX. Est.: 4h; GPU=True. Expected fidelity: quantitative match on reported error tables.

#4 — Lifex-cfd: an Open-source Computational Fluid Dynamics Solver for Cardiovascular Applications Method: finite element. Tests: poisson, navier-stokes, advection. Tools: FEniCS, OpenFOAM. Est.: 6h; GPU=False. Expected fidelity: quantitative match on reported error tables.

#5 — Enhancing training of physics-informed neural networks using domain-decomposition based preconditioning strate... Method: finite element. Tests: burgers, allen-cahn, advection. Tools: PyTorch or JAX. Est.: 6h; GPU=True. Expected fidelity: quantitative match on reported error tables.

#6 — Machine learning-accelerated computational fluid dynamics Method: finite element. Tests: navier-stokes, advection, diffusion. Tools: PyTorch, JAX. Est.: 6h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#7 — Dedalus: A flexible framework for numerical simulations with spectral methods Method: finite difference. Tests: poisson, navier-stokes, helmholtz. Tools: FEniCS, OpenFOAM. Est.: 6h; GPU=False. Expected fidelity: quantitative match on key figures.

#8 — Physics-informed neural networks for solving Reynolds-averaged Navier-Stokes equations Method: spectral. Tests: navier-stokes, stokes. Tools: PyTorch or JAX. Est.: 8h; GPU=True. Expected fidelity: quantitative match on key figures. Risks: no public code.

#9 — Learning in Modal Space: Solving Time-Dependent Stochastic PDEs Using Physics-Informed Neural Networks Method: spectral. Tests: burgers, navier-stokes, advection. Tools: PyTorch or JAX. Est.: 6h; GPU=True. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#10 — Poisson Flow Generative Models Method: spectral. Tests: poisson, diffusion. Tools: FEniCS, PETSc. Est.: 2h; GPU=False. Expected fidelity: quantitative match on key figures.

#11 — LNO: Laplace Neural Operator for Solving Differential Equations Method: neural network. Tests: heat equation, diffusion, reaction-diffusion. Tools: PyTorch or JAX. Est.: 6h; GPU=True. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#12 — Walk on Stars: A Grid-Free Monte Carlo Method for PDEs with Neumann Boundary Conditions Method: finite element. Tests: poisson, navier-stokes, helmholtz. Tools: FEniCS, PETSc. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#13 — Fast Poisson solvers for spectral methods Method: finite element. Tests: poisson, stokes. Tools: FEniCS, PETSc. Est.: 2h; GPU=False. Expected fidelity: quantitative match on reported error tables.

#14 — Deep Reinforcement Learning for Adaptive Mesh Refinement Method: finite element. Tests: poisson, schrödinger, wave equation. Tools: deal.II, AMReX. Est.: 8h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#15 — Godunov loss functions for modelling of hyperbolic conservation laws Method: finite difference. Tests: burgers, navier-stokes, diffusion. Tools: PyTorch or JAX. Est.: 5h; GPU=True. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#16 — Plane Wave Discontinuous Galerkin Methods for the 2D Helmholtz Equation: Analysis of the p-Version Method: finite element. Tests: helmholtz. Tools: FEniCS, Firedrake. Est.: 4h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#17 — Galerkin Approximations for the Stochastic Burgers Equation Method: finite element. Tests: burgers, heat equation, navier-stokes. Tools: SciPy, NumPy. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#18 — FLUPS: A Fourier-Based Library of Unbounded Poisson Solvers Method: spectral. Tests: poisson, helmholtz. Tools: FEniCS, PETSc. Est.: 2h; GPU=False. Expected fidelity: quantitative match on reported error tables.

#19 — Kernel-based active subspaces with application to computational fluid dynamics parametric problems using the d... Method: finite element. Tests: poisson, navier-stokes, advection. Tools: Clawpack, FEniCS. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables.

#20 — On the convergence of discontinuous Galerkin/Hermite spectral methods for the Vlasov-Poisson system Method: finite element. Tests: poisson, diffusion, vlasov. Tools: FEniCS, PETSc. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#21 — Improvements to the APBS biomolecular solvation software suite Method: finite element. Tests: poisson, diffusion. Tools: FEniCS, PETSc. Est.: 3h; GPU=False. Expected fidelity: quantitative match on key figures. Risks: no public code.

#22 — Fast Nonsymmetric Iterations and Preconditioning for Navier-Stokes Equations Method: finite element. Tests: navier-stokes, diffusion, stokes. Tools: FEniCS, OpenFOAM. Est.: 8h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#23 — New Laplace and Helmholtz solvers Method: finite element. Tests: helmholtz. Tools: FEniCS, Firedrake. Est.: 4h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#24 — Stability and error analysis of a class of high-order IMEX schemes for Navier-stokes equations with periodic b... Method: finite element. Tests: poisson, navier-stokes, stokes. Tools: FEniCS, OpenFOAM. Est.: 8h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#25 — Rate optimality of adaptive finite element methods with respect to the overall computational costs Method: finite element. Tests: poisson, diffusion. Tools: FEniCS, PETSc. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#26 — Comparison of Adaptive Multiresolution and Adaptive Mesh Refinement Applied to Simulations of the Compressible... Method: finite difference. Tests: specific test problem from paper. Tools: deal.II, AMReX. Est.: 6h; GPU=False. Expected fidelity: quantitative match on reported error tables.

#27 — Kinetic.jl: A portable finite volume toolbox for scientific and neural computing Method: finite volume. Tests: burgers, navier-stokes, advection. Tools: PyTorch or JAX, Julia. Est.: 6h; GPU=True. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#28 — Highly efficient and energy dissipative schemes for the time fractional Allen-Cahn equation Method: finite difference. Tests: poisson, allen-cahn, cahn-hilliard. Tools: FEniCS, PETSc. Est.: 5h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#29 — Modified Poisson-Nernst-Planck Model with Coulomb and Hard-sphere Correlations Method: finite difference. Tests: poisson, helmholtz, diffusion. Tools: FEniCS, PETSc. Est.: 3h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.

#30 — Optimized Schwarz Methods without Overlap for the Helmholtz Equation Method: krylov. Tests: helmholtz, wave equation. Tools: FEniCS, Firedrake. Est.: 4h; GPU=False. Expected fidelity: quantitative match on reported error tables. Risks: no public code.